

- (c) Saturn (d) Sun
(e) None of the above / More than one of the above.

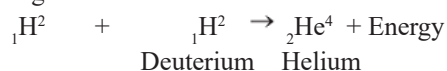
67th B.P.S.C. (Pre) (Re. Exam) 2022

Ans. (d)

The main celestial body of the solar system is Sun. The Sun is located at the center of solar system.

iii. The Sun

*The sun is the centre of our solar system. The bright outer layer that emits most of the radiation, particularly the visible light is known as the **Photosphere**. *According to NASA the diameter of the sun is **109.2** times bigger than our planet earth. The mean radius of the sun is 695700 km. Due to the very high temperature in the sun's core (appx. 15,000,000 °C) every matter present in it is either in a gaseous state or in the **plasma state**. *Plasma is the 4th state of matter after solid, liquid and gas. As per NASA, the surface temperature of the sun is **5500°C**. The entire source of the Sun's energy is a result of **nuclear fusion**- the fusing of hydrogen into helium, creating energy. *Nuclear reactions are of two types- **nuclear fusion** and **nuclear fission**. *In **nuclear fusion**, two or more atomic nuclei are combined to form one or more different atomic nuclei and subatomic particles (neutrons or protons). ***Nuclear fission** is a reaction in which a heavy nucleus splits spontaneously or on impact with another particle, with the release of energy. The release of energy from the stars is due to nuclear fusion. In stars when the temperature reaches between 1 to 1.5 crore Celsius. The atoms of hydrogen fuse together to form helium.



***Solar eclipse** is a situation when the complete or partial hiding of the sun is caused by the moon's passing between the sun and the earth. A solar eclipse occurs only on a **new moon day**. The outer layer of the sun which is visible during a solar eclipse is known as the **corona**. In any circumstances, the maximum duration of a total solar eclipse cannot exceed **7.5 minutes**. *When a solar flare enters the atmosphere of our planet, they it collides with the air particles and produces a beautiful display of a light show especially in the regions of upper latitude. This display is known as **Aurora lights**. At North Pole it is called **Aurora Borealis** and at the South Pole, it is called **Aurora Australis**. *A dark patch on the surface of the sun is known as a **sunspot**. Sunspots appear as regions with temperature about 1500K cooler than the surrounding photospheres. *We can often witness a white color ring around the sun and the moon, it is known as a

halo. It is caused when the light gets refracted or reflected due to the presence of cirrus and cirrostratus clouds in the upper layers of the atmosphere. *The earth is closest to the sun on the nearest date of **4th of January**. This phenomenon is known as **perihelion**, whereas on the nearest date **4th of July** the earth is at the farthest point from the sun and this phenomenon is called **aphelion**.

*Astronomical unit is used for measuring distance or length. For general reference, we can say that one Astronomical unit represents the mean distance between the earth and the sun (149.598×10^6 km or 149.6×10^6 km). In meters it is 149.59×10^9 ; In millimetres it is 149.59×10^{12} .

1. The substances present at the center of the sun are in:

- (a) Solid, liquid and gaseous states
(b) Liquid state only
(c) Gaseous state only
(d) Both liquid and gaseous state

U.P.P.C.S.(Pre) 2001

Ans. (*)

The temperature of the center of the the sun is very high (nearly 15000000° C) and because of such a high temperature, all the substances present at the centre of the sun are in the form of gas and plasma. Plasma is the fourth state of matter. Thus, none of the given answers is correct.

2. Solar energy is due to –

- (a) Ionization (b) Nuclear Fusion
(c) Nuclear Fission (d) Oxidation

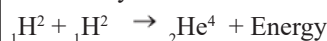
U.P.P.C.S. (Pre) 2006

U.P.P.C.S.(Pre) 2001

U.P.P.C.S. (Pre) 1996

Ans. (b)

The nuclear reaction may be divided into two categories namely, nuclear fusion and nuclear fission. In nuclear fission, one heavy nucleus divides into two or more nuclei while in nuclear fusion process two or more small nuclei combine together to form a heavy nucleus. Energy, heat and light released by stars are due to nuclear fusion.



3. Diamond Ring is a phenomenon observed :

- (a) At the start of a total solar eclipse
(b) At the end of a total solar eclipse
(c) Only along the peripheral regions of the totality trail
(d) Only in the central regions of the totality trail

I.A.S. (Pre) 1996

Ans. (c)

The diamond-ring effect occurs at the beginning and end of totality during a total solar eclipse. As the last bits of sunlight pass through the valley on the moon's limb and the faint corona around the sun is just becoming visible, it looks like a ring with a glittering diamond on it. Notably, the maximum duration of a total solar eclipse cannot be more than 7.5 minutes.

4. The solar eclipse occurs on -

- (a) Quarter Moon day (b) New Moon day
(c) Any day (d) Full Moon day (Poornima)

U.P.P.C.S. (Pre) 1991

Ans. (b)

A solar eclipse occurs when the Moon comes between Earth and the Sun, and the Moon casts a shadow over the Earth. A solar eclipse can only take place at the phase of New Moon (Amavasya), when it passes directly between the Sun and the Earth and its shadow falls upon the Earth's surface.

5. Every solar eclipse occurs on -

- (a) Poornima only (b) Amavasya only
(c) (a) and (b) (d) Neither (a) nor (b)

U.P. U.D.A./L.D.A. (Pre) 2006

R.A.S./R.T.S. (Pre) 2000

R.A.S./R.T.S. (Pre) 1999

Ans. (b)

See the explanation of the above question.

6. When does solar eclipse occur ?

- (a) When the sun comes between the earth and the moon.
(b) When the earth comes between the sun and the Moon.
(c) When the moon comes between the earth and the sun
(d) None of the above

47th B.P.S.C (Pre) 2005

M.P.P.C.S. (Pre) 1993

U.P.P.C.S. (Pre) 1990

Ans. (c)

See the explanation of the above question.

7. The Solar Eclipse achieves totality only in limited geographical regions because -

- (a) The size of the shadow of the Moon on the Earth is small as compared to the cross section of the Earth.
(b) The Earth is not a smooth flat surface but has elevations and depressions.
(c) The trajectories of the Earth around the Sun and the Moon around the Earth are not perfect circles.
(d) Sun rays can reach most of the peripheral regions of the shadow of the Moon due to atmospheric refraction.

I.A.S. (Pre) 1993

Ans. (a)

A solar eclipse occurs when the moon comes between the earth and Sun. In this situation, the part of the earth that is towards the Sun receives the shadow of the Moon and also the Moon's shadow falling on the Earth is smaller than the cross-section of the Earth. This is the reason that the total or full Solar eclipse is not visible on every part of the Earth together, but visible only to a limited portion of the Earth.

8. Sun's halo is produced by the refraction of light in -

- (a) Water vapour in stratus clouds
(b) Ice crystals in Cirro-Cumulus clouds
(c) Ice crystals in Cirrus clouds
(d) Dust particles in Stratus clouds

I.A.S. (Pre) 2002

Ans. (c)

The Sun's halo is a circle that appears around the sun. Whenever Sun rays pass through the ice crystals present at the upper level of cirrus clouds the sun's halo becomes visible due to the refraction of rays.

9. One Astronomical unit is average distance -

- (a) Between Earth and Sun
(b) Between Earth and Moon
(c) Between Jupiter and Sun
(d) Pluto and Sun

Jharkhand P.C.S. (Pre) 2011

I.A.S. (Pre) 1998

Ans. (a)

An astronomical unit is the mean distance between the Earth and the Sun. $1 \text{ AU} = 149.598 \times 10^6 \text{ km}$ ($150 \times 10^6 \text{ km}$) or 149.6 million km. In other various units its value is as follows.

In Metre - $1 \text{ AU} = 149.59 \times 10^9$

In millimetre - $1 \text{ AU} = 149.59 \times 10^{12}$

10. The average or mean distance between the earth and the sun is -

- (a) $70 \times 10^5 \text{ km}$ (b) $100 \times 10^5 \text{ km}$
(c) $110 \times 10^6 \text{ km}$ (d) $150 \times 10^6 \text{ km}$

I.A.S. (Pre) 1997

Ans. (d)

See the explanation of the above question.

11. The distance between the earth and the sun is -

- (a) 107.7 Million km (b) 142.7 Million km
(c) 146.6 Million km (d) 149.6 Million km

U.P.P.C.S. (Pre) 2010

Ans. (d)

See the explanation of the above question.